Course Information

ME 46300: Engineering Design

Application of the design process to the design of various engineering components and systems. Mathematical modeling in design is emphasized. Design problems from all areas of mechanical engineering are considered.
3 credit hours. Face-to-face.

CRN-Section	54528-011	20175-013	20176-014	11509-020
Class Type	Lecture	Lab	Lab	Lab
Day	M/W	T/Th	T/Th	W/F
Time	9:30 am – 10:20 am	8:30 am – 10:20 am	10:30 am – 12:20 pm	1:30 pm – 3:20 pm
Location	WTHR 104	ME 1185		
Instructor		Euiwon Bae	Eric Holloway	Beth Hess
E-mail		ebae@purdue.edu	eahollow@purdue.edu	bethhess@purdue.edu

Learning Resources, Technology and Texts

All course materials will be available on Brightspace.

Learning Outcomes

By the end of the course, you will:

1. Be able to discuss and articulate the viewpoint/perspective that engineering design problems are open-ended and multifaceted.
2. Have experienced a collaborative design methodology.
3. Have strengthened and refined skills in leadership, teamwork, communication, project planning, innovation, design and entrepreneurship.
4. Have applied core course materials, engineering design methods, and testing to practical open-ended design problems.
5. Have reinforced a philosophy of professional and ethical behavior.
6. Have applied Standards to open-ended design, analysis, build, and test problems.
7. Been provided with a practical foundation for knowledge acquisition and continued learning as needed in your future career.

Fall 2025 Schedule Rev. A

Senior Design ME46300

PURDUE UNIVERSITY
Mechanical Engineering

Updated: 31-July-2024

Legend for Lab Schedule	
PDR - Preliminary Design Review	
CDR - Critical Design Review	**SR** - Safety Review
FDR - Final Design Review	**PE** - Peer Evaluation

Lecture Meeting Time & Location: M/W 9:30-10:20am WHTR 104

Lab Meeting Location: PEARL Lab. Days & times per Lab section

	LECTURE SCHEDULE					LAB SCHEDULE				
	Date	Day	#	Lecturer	Topic	Date	Day	Assignments Due	Assignment Submission	Student Presentations
1	8/25	M	1	Hess	**Course Introduction | Applying Design Process to ME463 | Schedule**					
	8/27	W	2	Hess	**Team exercise - Soft skills**	8/26 or 8/27	T or W		None	Prob. Def., Stakeholders
						8/28 or 8/29	R or F	**Teams & Projects set and review (by instructor)**	N/A	Mission and Vision
2	9/1	M			**LABOR DAY - NO CLASS**					
	9/3	W	3	Bae	**Report | Presentation|Basic Phyiscs Model**	9/2 or 9/3	T or W	**Informed Consent, Team Registration**	Brightspace (BS), BS	Team exercise
						9/4 or 9/5	R or F	**Concept Generation activity in class**		Charter, Market Analysis
3	9/8	M	4	Montalvo	**Decision Matrix|Design Constraints**					
	9/10	W			**Industrial Roundtable - NO CLASS**	9/9 or 9/10	T or W			Concept generation and down selection
						9/11 or 9/12	R or F			Concept generation and down selection
4	9/15	M	5	Hess	**Prelim. Prototype | Facilities | resource scavenger hunt**					
	9/17	W			**PDR review and preparation - NO CLASS**	9/16 or 9/17	T or W			PDR slides review
						9/18 or 9/19	R or F	**Oral PDR, Written PDR, SR #1**	BS, BS, email Sherwood	Oral PDR presentation
5	9/22	M	6	Montalvo	**GD&T 1 | Intro for Paraellel Track Video Lectures**					
	9/24	W	7	Montalvo	**GD&T 2**	9/23 or 9/24	T or W			Preliminary Prototype scope/design
						9/25 or 9/26	R or F	Prelim. proto >110V require Sherwood's approval		Preliminary Prototype scope/design
6	9/29	M	8	Hess/Bae	**IP Law overview | Patents**					
	10/1	W	9	Hess	**Shape Synthesis | Purchasing**	9/30 or 10/1	T or W	**Preliminary Prototype Demo**		In class demo and presentation slides
						10/2 or 10/3	R or F			Design, Schedule, Risk review
7	10/6	M	10	Bae	**Validation | Testing**					
	10/8	W	11	Hess	**FE Exams |MBA Pathways**	10/7 or 10/8	T or W			Design, Schedule, Risk review
						10/9 or 10/10	R or F			Design, Schedule, Risk review
8	10/13	M			**OCTOBER BREAK - NO CLASS**					
	10/15	W			**CDR review and preparation - NO CLASS**	10/14 or 10/15	T or W			Oral CDR review
						10/16 or 10/17	R or F			Oral CDR presentaton
9	10/20	M			**No Lecture**					
	10/22	W			**No Lecture**	10/21 or 10/22	T or W	**Oral CDR, Written CDR, SR#2, PE#1**	BS, BS, email Sherwood, CATME	
						10/23 or 10/24	R or F	**All Final Prototype purchases placed**	Email (purchase requests)	
10	10/27	M			**No Lecture**					
	10/29	W			**No Lecture**	10/28 or 10/29	T or W			
						10/30 or 10/31	R or F			
11	11/3	M			**No Lecture**					
	11/5	W			**No Lecture**	11/4 or 11/5	T or W			Prototype fab., Schedule, Risk review
						11/6 or 11/7	R or F			
12	11/10	M			**No Lecture**					
	11/12	W			**No Lecture**	11/11 or 11/12	T or W	**SR #3** (must be submitted & approved before final prototype testing)	email M. Sherwood & Instructor	Prototype fab., Validation plan
						11/13 or 11/14	R or F			
13	11/17	M			**No Lecture**	11/17	M	**Malott Project Descriptions**	BS	
	11/19	W	12	Hess	**Course Wrap-up & End-of-semester Schedule**	11/18 or 11/19	T or W			Prototype fab., Schedule, Risk review
						11/20 or 11/21	R or F	**Final Prototype check 1**		Initial Final Prototype
14	11/24	M			**No Lecture**					
	11/26	W			**THANKSGIVING VACATION - NO CLASS**	11/25 or 11/26	T or W			Validation results, Design iterations
						11/27 or 11/28	R or F	**THANKSGIVING VACATION - NO Lab**		
15	12/1	M			**No Lecture**	12/1	M	**Malott poster (by 8am Monday)**	Email Mike Black (blackm@purdue.edu)	
	12/3	W			**No Lecture**	12/2 or 12/3	T or W			
						12/4 or 12/5	R or F	**Final Prototype check 2**		Final Prototype Demo
16	12/8	M			**No Lecture**					
	12/10	W			**No Lecture**	12/9 or 12/10	T or W	**Oral FDR, Written FDR, PE#2**	Present & BS, BS, CATME	Oral FDR
						12/11 & 12/12	R & F	**Malott Innovation Awards Event**		Malott - No Lab
Finals	12/15	M								
	12/17	W				12/16	T	**Clean up PEARL lab rooms (9:00am - 12:00pm)**		

Assignments

Phase	Deliverable	Points
Preliminary Design Review (PDR)	Team Registration and Charter	50
	Lecture and Lab Attendance	50
	Oral PDR presentation	50
	Written PDR report	50
	Safety Review #1 (SR1)	0
Critical Design Review (CDR)	Preliminary Prototype Review	100
	Oral CDR presentation	100
	Written CDR report	100
	Safety Review #2 (SR2)	0
	Peer Evaluation #1 (PE1)	100
Final Design Review (FDR)	Safety Review #3 (SR3)	0
	Final Prototype Review	0
	Oral FDR presentation	150
	Written FDR report	150
	Peer Evaluation #2 (PE2)	100
	Course and Instructor Evaluations	10 (bonus)
	End-of-semester workstation clean-up	Up to 50 (deduction)
	Total	**1000**

Notes about Assignments

- Due dates and times are set and managed via your Lab instructor and Lab Brightspace site. See Course Schedule for typical due dates.
- All assignments are submitted via your Lab Brightspace page or per instructions provided by your Lab instructor. Late assignments will result in point deductions per the discretion of your Lab instructor.
- When applicable, rubrics will be provided for assignments. All assignments are graded by your Lab instructor. Questions about grading or rubrics should be addressed to your Lab instructor.
- All assignments except PE1, PE2 and attendance are team grades. Every team member receives the same grade. Exceptional circumstances will be reviewed by your Lab instructor and grades may be adjusted for individual students at your Lab instructor's discretion.

Grading Scale

Final grades will be calculated by adding the total points earned and translating those numbers (out of 100) to letter grades. The grade cutoffs are the following.

A+ :	970-1000	A :	930-969	A- :	900-929
B+ :	870-899	B :	830-869	B- :	800-829
C+ :	770-799	C :	730-769	C- :	700-729
D+ :	670-699	D :	630-669	D- :	600-629
F :	<600				

The final date to withdraw from a course with a W for Fall 2025 is November 25.

Attendance Policy

This course follows Purdue's academic regulations regarding attendance, which states that students are expected to be present for every meeting of the classes in which they are enrolled.

Under academic regulations, excused absences may be granted by the Office of the Dean of Students (ODOS) for cases of grief/bereavement, military service, jury duty, parenting leave, or certain types of medical care. For absences that fall under excused absence regulations, you or your representative should contact ODOS or go to the ODOS website to complete appropriate forms for instructor notification.

Excused absences may also be granted for University-sponsored activities and religious observations.

When conflicts or absences can be anticipated, please inform your instructor of the situation as far in advance as possible.

Late assignment submissions will not be accepted unless the assignment submission falls on the date of an excused absence.

Academic Integrity

Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breaches of this value by either emailing integrity@purdue.edu or by calling 765-494-8778. While information may be submitted anonymously, the more information is submitted the greater the opportunity for the university to investigate the concern. More details are available on the Academic Resources table on the Brightspace homepage.

Nondiscrimination Statement

Details on Purdue's full Nondiscrimination Policy Statement are available on the Academic Resources table on the Brightspace homepage.

Accessibility

Purdue University strives to make learning experiences accessible to all participants. If you anticipate or experience physical or academic barriers based on disability, you are encouraged to contact the Disability Resource Center (DRC) as soon as possible at drc@purdue.edu or by calling 765-494-1247.

If the DRC has determined reasonable accommodations that you would like to utilize in class, you must release your Course Accommodation Letter (CAL) to your instructor (instructions on sharing your CAL: https://www.purdue.edu/drc/students/course-accommodation-letter.php). You are strongly encouraged to contact your instructor as soon as possible to discuss implementation of your accommodations.

Mental Health and Wellness

If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try Therapy Assistance Online (TAO), a web and app-based mental health resource available courtesy of Purdue Counseling and Psychological Services (CAPS)**.** TAO is available to students, faculty, and staff at any time.

If you need support and information about options and resources, please contact ODOS.

If you find yourself struggling to find a healthy balance between academics, social life, stress, etc., sign up for free one-on-one virtual or in-person sessions in West Lafayette with a Purdue Wellness Coach at RecWell. Student coaches can help you navigate through barriers and challenges toward your goals throughout the semester. Sign up is free and can be done on BoilerConnect. Students in Indianapolis will find support services curated on the Vice Provost for Student Life website.

If you're struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact CAPS at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS offices in West Lafayette or Indianapolis.

The Community, Assistance and Resources for Engineering Students (CARES) Hub exists to promote and enhance the well-being of all Purdue engineering students. The **CARES Hub** offers study spaces, an on-site therapist, well-being workshops, social events, peer mentoring, and more. Check out the CARES Hub website for more information.

Emergency Preparedness

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructors or TAs via email or phone. You are expected to read your @purdue.edu email on a frequent basis.

See Purdue's Information on Emergency Preparation and Planning. This website covers topics such as Severe Weather Guidance, Emergency Plans, and a place to sign up for the Emergency Warning Notification System. You are encouraged to download and review the Emergency Preparedness for Classrooms document.

The **Emergency Preparedness plan for the classroom** will be shared during the first day of class. Please make note of items like:

- The location to where we will proceed after evacuating the building if we hear a fire alarm.
- The location of our Shelter in Place in the event of a tornado warning.
- The location of our Shelter in Place in the event of an active threat such as a shooting.